

Week 12 Cheat Sheet: RL, MARL, and Stochastic Games

1. Deterministic vs Stochastic Systems

Deterministic:

$$x_{t+1} = f(x_t, u_t)$$

Stochastic (with Nature θ):

$$x_{t+1} = f(x_t, u_t, \theta_t)$$

Transition Probability:

$$P(x_{t+1} \mid x_t, u_t)$$

Proposal Connection: Other agents behave like uncertainty (“Nature”). We must reason probabilistically about their actions.

2. Bellman Equation (Core Principle)

$$J^*(x_t) = \min_{u_t} \mathbb{E} [l(x_t, u_t) + J^*(x_{t+1})]$$

Meaning: Optimal decision = immediate cost + expected future cost.

Proposal Connection: Robots balance short-term safety vs long-term efficiency.

3. Value vs Policy Iteration

Value Iteration:

$$J(x) \leftarrow \min_u \left[l(x, u) + \sum_{x'} P(x'|x, u) J(x') \right]$$

Policy Iteration:

- Evaluate current policy
- Improve policy greedily

Proposal Connection:

- Greedy = local decisions only
- Global lookahead = iterative planning

4. Reinforcement Learning (RL)

Phases:

1. Learning dynamics
2. Planning policy
3. Execution

Proposal Connection: Robots can learn interaction patterns instead of relying only on fixed rules.

5. Temporal Difference (TD) Learning

$$J(x_t) \leftarrow J(x_t) + \alpha(l(x_t, u_t) + J(x_{t+1}) - J(x_t))$$

Idea: Update using partial information.

Proposal Connection: Online updates for real-time decision making.

6. Q-Learning

$$Q^*(x, u) = l(x, u) + \mathbb{E} \left[\min_{u'} Q^*(x', u') \right]$$

$$\pi(x) = \arg \min_u Q(x, u)$$

Meaning: Evaluate each action directly.

Proposal Connection: Robots compare actions like “wait vs move”.

7. Deep Reinforcement Learning

$$Q(x, u) \approx Q(x, u; \omega)$$

Idea: Use neural networks for large state spaces.

Proposal Connection: Multi-robot environments are continuous $\rightarrow$ require approximation.

8. Stochastic Games (Multi-Agent Setting)

$$\langle N, S, A_i, P, R_i, \gamma \rangle$$

Key Idea: Each agent’s reward depends on others.

Proposal Connection: Your problem is inherently a stochastic game.

9. Multi-Agent RL (MARL)

Value-based: Extend Q-learning

Policy-based:

$$\nabla_{\theta_i} J_i = \mathbb{E} [\nabla \log \pi_i \cdot Q_i]$$

Challenge: Joint action space explosion.

Proposal Connection: Robots must coordinate under complexity.

10. Nash Equilibrium

$$\pi_i^* \in \text{BestResponse}(\pi_{-i})$$

Definition: No agent benefits from unilateral deviation.

Complexity:

- PPAD-hard
- NP-hard
- PSPACE-hard

Proposal Connection: Ideal coordination is too expensive $\rightarrow$ need approximations.

11. ORCA (Optimal Reciprocal Collision Avoidance)

Core Idea:

- Each robot independently selects a velocity
- Avoids collisions using geometric constraints
- No communication required

Velocity Obstacle (VO): Set of velocities that would lead to collision.

ORCA Constraint: Each robot chooses velocity from a safe half-plane.

Optimization:

$$v_{\text{new}} = \arg \min_{v \in \text{safe set}} \|v - v_{\text{pref}}\|$$

Proposal Connection:

- ORCA = greedy, local, reactive baseline
- No negotiation or global reasoning
- Your work extends beyond ORCA using coordination and arbitration

Big Picture

- Problem = stochastic multi-agent decision making
- ORCA = fast but local
- Game theory = optimal but expensive
- Your proposal = bridge via negotiation + LLM arbitration